DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

ANO 2025/2026

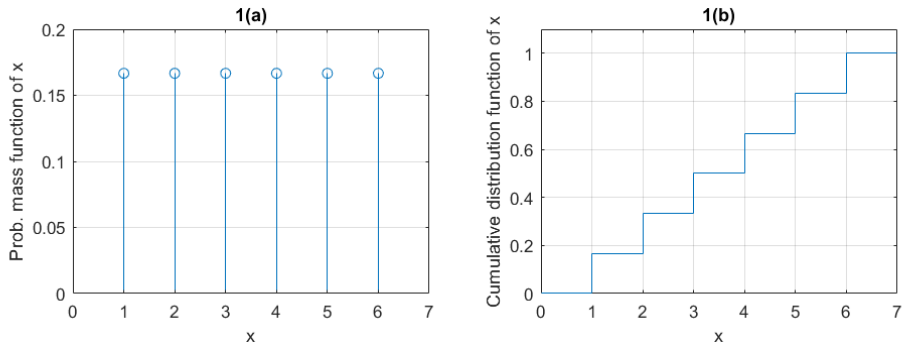
MÉTODOS PROBABILÍSTICOS PARA ENGENHARIA DE COMPUTADORES E INFORMÁTICA

PRACTICAL GUIDE NO. 2

1. Consider the random variable X corresponding to the face that is up on the roll of a die. Using the theoretical values, generate in Matlab the following two graphs in the same figure:

(a) a graph with the probability mass function of X (use Matlab *stem* function),

(b) a graph with the cumulative distribution function (use Matlab *stairs* function).



2. Consider a box containing 90 banknotes of 5 euro, 9 banknotes of 10 euros and 1 banknote of 100 euros.

(a) Describe the sampling space of the random experiment of taking at random a banknote out of the box, and the probabilities of the elementary events.

(b) Now consider the random variable X representing the value of the banknote taken randomly from the box. Describe the sampling space and the probability mass function of X.

(c) Determine the cumulative distribution function of X and plot it in Matlab.

3. Consider the toss of 4 balanced coins. Let X be the random variable representing the number of heads observed in the 4 tosses.

(a) Estimate by simulation the probability mass function, $p_X(x)$, of the random variable X.

(b) Estimate the expected value, variance, and standard deviation of X based on $p_X(x)$.

(c) Identify the type of distribution of X and write the theoretical expression of its probability mass function.

(d) Calculate the theoretical values of the probability mass function of X and compare them with the values estimated by simulation in (a).

(e) Calculate the theoretical values $E[X]$ and $\mathrm{Var}[X]$ and compare them with the values obtained in (b).

(f) Based on the theoretical values of the probability mass function of X, calculate:

i. the probability of obtaining at least 2 heads,

ii. the probability of obtaining up to 1 head,

iii. the probability of obtaining between 1 and 3 heads.

4. (Optional) Knowing that a manufacturing process produces 30% defective parts and considering the random variable X representative of the number of defective parts in a random sample of 5 parts:

(a) Estimate by simulation:

i. the probability mass function of X,

ii. the graph representing the cumulative distribution function of X,

iii. the probability that at most 2 of the parts in a sample are defective.

(b) Compute theoretically:

i. the graph representing the cumulative distribution function of X,

ii. the probability that at most 2 of the parts in a sample are defective.

5. (Optional) Assume that each engine of an airplane may fail with probability p and that the failures are independent between engines. Assume also that the plane crashes if more than half of the engines fail. Under these assumptions, do you prefer to fly in a plane with 2 engines or 4 engines? Use the distribution that is most appropriate.

Suggestions: You have at least 2 alternatives: (1) get expressions for the probability of each type of plane crashing as a function of p and use the quotient between them to answer the question, (2) perform the calculations for a set of concrete values of p (for example: *p= logspace(-3,log10(1/2),100))* and use a graph simultaneously showing the crashing probability of each type of plane.

6. The binomial distribution tends to the Poisson distribution with rate λ when $n \to +\infty$, $p \to 0$ and $\lambda = np$ constant [1]. Consider a data link with a BER (bit error rate) equal to 10^{-5} (the errors in the different bits of a data packet are statistically independent). Compute theoretically the following probabilities based on the binomial distribution and on the Poisson distribution:

(a) the probability of a data packet of size 100 Bytes to be received without errors,

(b) the probability of a data packet of size 1000 Bytes to be received with at least 2 errors.

7. Assume that the number of email messages received by an email server has a Poisson distribution with an average of 5 messages per second. Determine theoretical and estimate by simulation (in this case, use Matlab *poissrnd* function to generate random values with the Poisson distribution):

(a) the probability of the server to receive less than 10 messages in an interval of 2 seconds,

(b) the probability of the server to receive more than 30 messages in an interval of 5 seconds.

8. Consider a random variable X with an exponential distribution with rate $\lambda = 1$:

(a) Using Matlab *exppdf* function, plot the probability density function of X for the values from 0 to 10.

(b) Using Matlab *exprnd* function, generate a vector with 10^5 random values of X and plot a histogram of the generated values with 100 bins. Is the histogram in accordance with the probability density function of X?

9. Consider a random variable X with an exponential distribution with rate $\lambda = 0.2$. Start by generating a vector with 10^5 random values of X.

(a) Show that the average of the generated values is in accordance with the average of the exponential distribution.

(b) With the 10^5 random values generated, show that the exponential distribution has no memory, i.e., $P(X > s + t \mid X > t) = P(X > s)$. For example, use the values $s = 2$ and $t = 3$ (but the same happens for any pair of values).

[1] A random variable X with a Poisson distribution with rate λ has the probability mass function $P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$.

10. Consider a random variable X with an exponential distribution with rate λ_X and a random variable Y with an exponential distribution with rate λ_Y. Start by generating a vector with 10^5 random values of X (considering $\lambda_X = 0.4$) and a vector with 10^5 random values of Y (considering $\lambda_Y = 0.1$). With these values, show that $P(X < Y) = \frac{\lambda_X}{\lambda_X + \lambda_Y}$.

11. (Optional) Consider that the number of typographical errors on each page of a book is a random variable with a Poisson distribution with $\lambda = 0.02$. Determine the probability of a book of 100 pages to have at most 1 typographical error.

12. Consider that the number of students in a course unit is a random variable X with a Gaussian distribution (average 100 students and standard deviation 10 students). The course unit has practical classes with a maximum number of 20 students per class, and each student is enrolled in one practical class. Consider the probability that n classes are enough to enrol all students.

(a) Using Matlab *randn* function, estimate by simulation (based on 10^5 random values of X) the probability that n classes are enough (for n = 5, 6 and 7).

(b) Using Matlab *normcdf* function, compute the theoretical probability that n classes are enough (for n = 5, 6 and 7), and compare them with the simulation results.

13. A factory produces cookies, and the weight of each cookie is a random variable X_i. Assume the following:

- The cookie weights are i.i.d. (independent and identically distributed)
- Mean weight: μ= 20 g
- Variance: $\sigma^2 = 9\ \mathrm{g}^2$

Let $S_n = \sum_{i=1}^{n} X_i$ be the total weight of a batch of n cookies.

(a) Compute theoretically $E[S_n]$ and $\mathrm{Var}[S_n]$

(b) Using only the mean, give an upper bound on the probability that a single cookie weighs at least 30 g.

(c) Using both the mean and variance, bound the probability that a single cookie deviates from its mean weight by more than 10 g.

(d) Estimate the probability that the total weight, S_{100}, is between 1970 g and 2030 g.